

Week 8 Tutorial Session: Tutorial 5

Instruction:

Submit your answer for this tutorial to Spectrum by the deadline. Your submission should be in the form of a .docx file. State your name, matric number (for example 12345678/9) and tutorial group in your file. Be prepared to share your answer in the lecture/tutorial session.

You are not allowed to use any AI tools in any way. Please refer to relevant lecture slides (uploaded under Week 7 Lecture in Spectrum) yourself, understand them yourself and write/draw your answers yourself. Please don't neglect to use and train your human intelligence. Nevertheless, you can use software tools to draw the diagrams yourself.

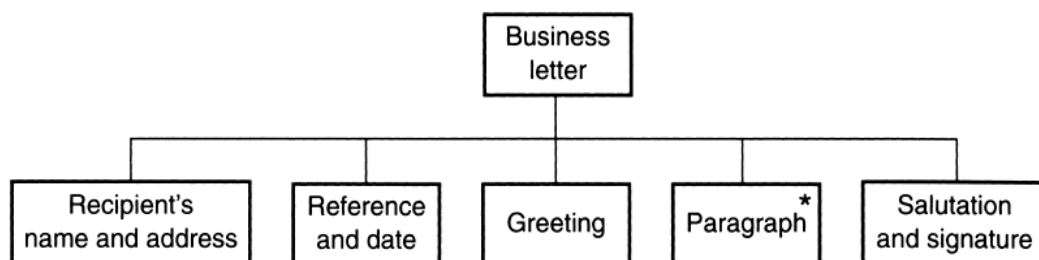
Answer all the questions.

1. A salary payment system computes the monthly net salary of an employee, prints the monthly salary slip for the employee, and generates the instruction to deposit the employee's bank account with the net salary. To support its operation, this system uses three data stores or files:
 - a) employees data store which contains the details of each employee such as the employee name, id, gross salary and bank account number;
 - b) tax payment data store that contains information on tax payable based on the ranges of the gross salary;
 - c) salary data store that contains the details of salary payment of each month for each employee.

The system calculates the monthly net salary by deducting the tax payable from the gross salary. The system obtains the employee id and the month for which the net salary is to be calculated from the user of the system.

Draw a Level 0 and a Level 1 Data Flow Diagrams (DFDs) for this system.

2. The following are Jackson structure diagrams. Why are these diagrams wrong? Redraw them using the correct forms.
 - (a)



(b)

